# Module 4

# THERMODYNAMICS

## THERMODYNAMICS

- **Branch of science which deals with heat and work.**
- Greek words therme (heat) and dynamics (power)

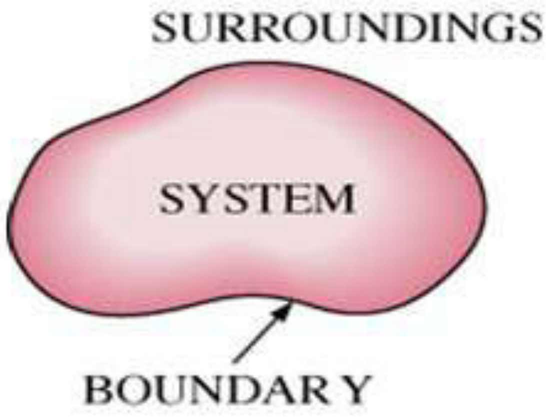


## Thermodynamic System

- **Thermodynamic System** or **System** is a definite area or a space where some thermodynamic activity is taking place.

Or

- It is the area where the study is focussed on.






**SURROUNDING**

- The matter or region outside the system is termed as surrounding.

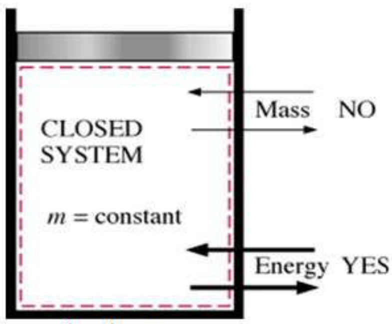
**BOUNDARY**

- The real or imaginary envelope which encloses the system and separates with surrounding.



**Closed System**

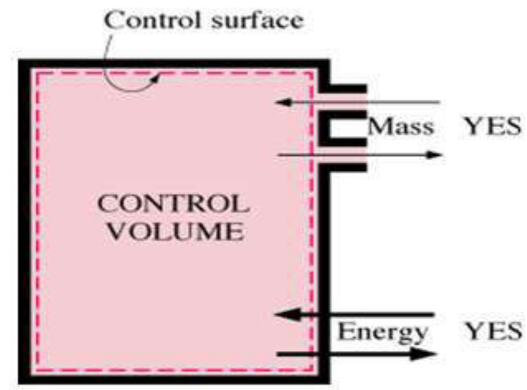
- System containing some matter in which there is no transfer of mass across its boundary.




Closed system

**Open System**

➢ Mass and energy can cross the boundary of a Control Volume




Open system or Control Volume



**Isolated System**

➢ There will not be any mass or energy transfer across the boundary.



**Macroscopic Approach**

➢ The macroscopic approach to thermodynamics is concerned with the gross or overall behaviour.

➢ Certain quantity of matter is considered with out taking to account the events occurring at the molecular level.

## Microscopic Approach

- The approach considers that the system is matter of a very large number of discrete particles known as molecules.

- Large number of variables are needed to describe a system. So approach is complicated



## STATE

- The exact condition of a system described by its properties is called its state.

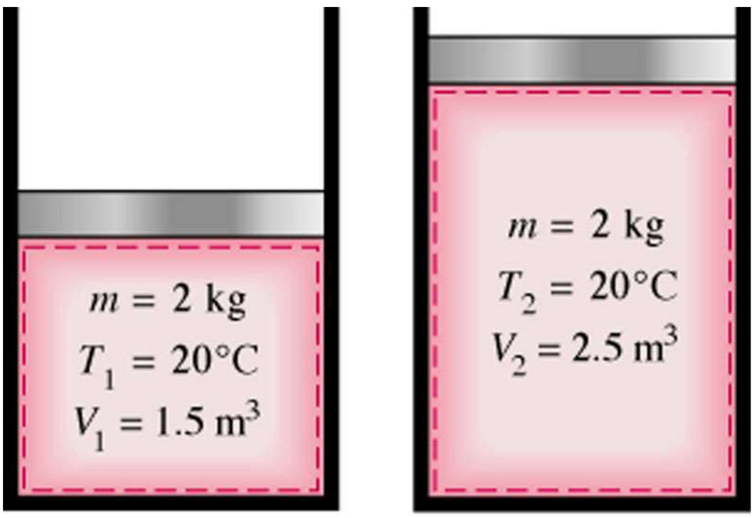
  ***If the value of one property changes, the state will change to a different one.***

## PROPERTIES

- The variables, which determine the state, are called as its properties.

  *E.g.: Volume, temperature, pressure, enthalpy, entropy*
  *The fundamental properties are pressure, volume and temperature*.






(*a*) State 1 (*b*) State 2

## EXTENSIVE PROPERTY:

- System whose value is equal to sum of their individual parts.

***eg. Total Mass( Total Mass of a body is equal to the summation of mass of many individual parts of the same system), volume etc.***

- property proportional to mass of the system.

## INTENSIVE PROPERTY:

- System whose value for the entire system is not equal to sum of their individual parts
- Property independent of mass of system.

***eg. Temperature (Temperature of a system is not equal to sum of the temperature of its individual parts), Pressure etc.***

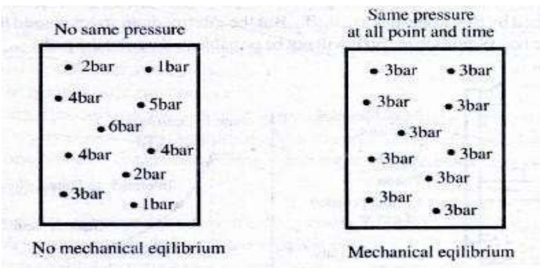


# Thermodynamic equilibrium

- Balance of mechanical, thermal (no variation in Temperature), and chemical equilibrium.

**Mechanical Equilibrium**:

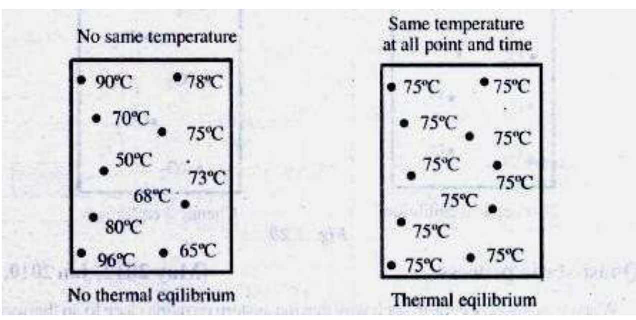
- A system is said to be in **Mechanical Equilibrium** when there is no unbalanced force acting on any part of the system or the system as a whole.






# Thermal Equilibrium:

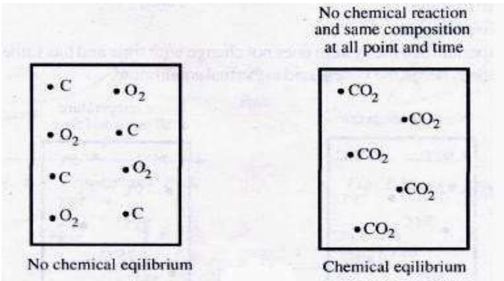
- A system is said to be in **Thermal Equilibrium** when there is no temperature difference between the parts of the system or between the system and the surroundings.



## Chemical Equilibrium :

- A system is said to be in **Chemical Equilibrium** when there is no chemical reaction with in the system,

  and also there is no movement of any chemical constituent from one part of the system to the other.



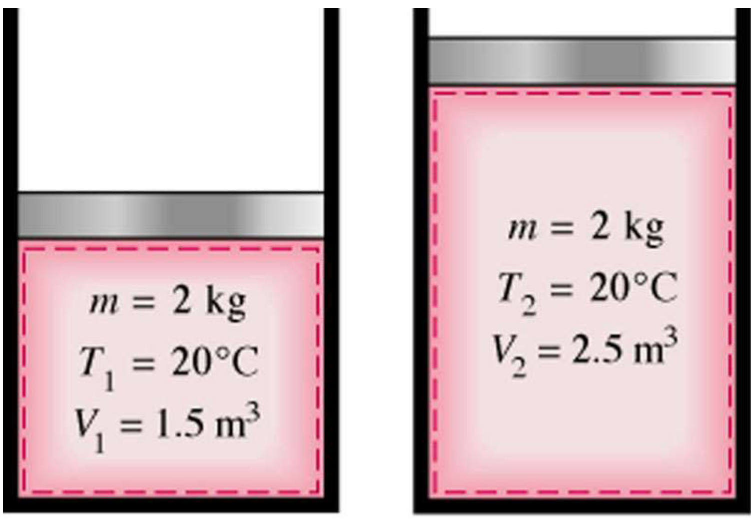



## PROCESS

- **A process is a transformation from one state to another.**

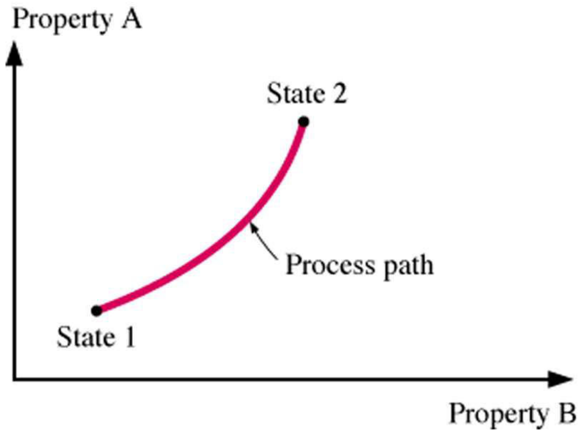
- *When any of the properties of a system change, the state changes and the system is said to have undergone a process*.






(*a*) State 1 (*b*) State 2

# PROCESS






## Thermodynamic Cycle .

- At the end of the last process if the system returns to its original state, it is said to have completed one **Thermodynamic Cycle** or **Cyclic Process**.

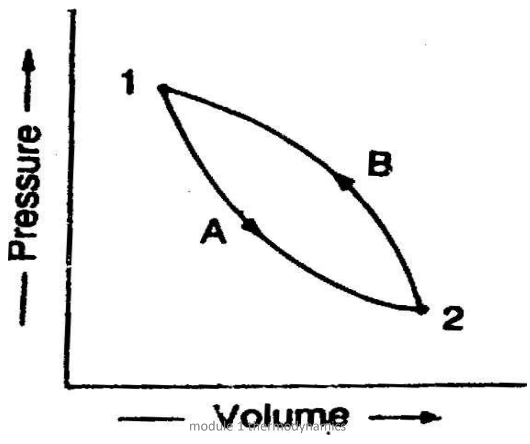
**OR**

- When process or processes are performed on a system in such a way that the final state is identical with initial state, it is known as **Thermodynamic Cycle** or **Cyclic Process**.



## **Thermodynamic Cycle** or **Cyclic Process**.

*Cycle. **1-A-2** and **2-B-1** are processes and **1-A-2-B-1** is a Thermodynamic Cycle or Cyclic Process.*



## Quasi-Static or Quasi-Equilibrium Process

## or

## Reversible Process

- In a process if the system deviation from thermodynamic equilibrium is very small, then the process is known as **Quasi-Static** or **Quasi-Equilibrium Process or Reversible Process**

**OR**

- Process which can be reversed in direction and the system traces same equilibrium states

- It is represented by continuous line on property diagram



## Irreversible Process

## or

## Non-Equilibrium Process

- If the system deviation from Thermodynamic Equilibrium is very large in a process, then the process is known as **Irreversible Process** or **Non-Equilibrium Process**

- Represented by broken lines in the property diagram



## Path of the Process

- The series of states through which a system passes during a process is called **Path of the Process**

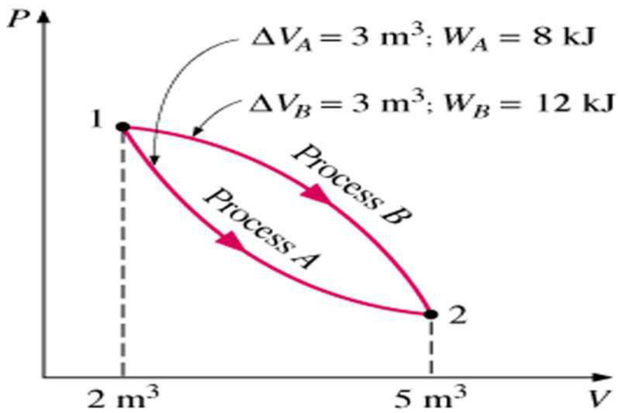
## Path function

- If the value of a thermodynamic variable depends upon the path followed in going from one state to another, then the variable is a **path function.**

## State Functions or point functions.

- If the value of a thermodynamic variable is independent of the path followed it is called **State Functions** or **point functions**.






At state 2 the value of pressure volume and temperature will be the same whether the state 2 is arrived along the path 1 – A – 2 or 1 – B – 2 i.e., the value of pressure, volume and temperature at state 2 is independent of the path followed and depends only on the state 2.



## Perfect or ideal gas

- Gas which strictly obeys all the gas laws under all conditions of pressure and temperature.
- In fact no actual or real gas, which exists in nature, is perfect.
- ***In engineering applications gases such as air, nitrogen, hydrogen etc, are considered as perfect gases.***

# GAS LAWS

***Boyle's law*** and ***Charles' law*** govern the behavior of a perfect gas.

## Boyle's law:

- The absolute pressure of a given mass of a perfect gas varies inversely to its volume when the temperature remains constant.

$$P \,\alpha\, \frac{1}{V}$$

PV=constant

$$p_1v_1 = p_2v_2 = p_3v_3 = pv = Constant$$, **at constant temperature**



## Charles's law:

- The volume of a given mass of a perfect gas varies directly to its absolute temperature when the pressure remains constant.

$V \,\alpha\, T$ when P=constant

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$ **, when pressure remains constant**



## Characteristic Gas Equation

$$P \,\alpha\, \frac{T}{V}$$

$$\frac{PV}{T} = \text{Constant}, \quad \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$PV = mRT$ , R is Characteristic gas constant

$PV = nR_uT$ , $R_u$ is Universal gas constant

$R_u = 8314$ J/kgK

R for air = 287 J/kgK

# Heat

- It is the energy transfer across the boundary of a system due to temperature difference between system & Surroundings. It is denoted by Q.
- The transfer of heat into a system is frequently referred to as heat addition and the transfer of heat out of a system as heat rejection.
- H.T Mechanisms :
  Conduction
  Convection
  Radiation



## Latent heat

- **The amount of heat added to or removed from a substance to produce a change in phase**.
- When latent heat is added, no temperature change occurs.

## Sensible heat

- **The amount of heat added to or removed from a substance which produces a change in temperature**



## Specific heat

- The specific heat of a substance is defined as the amount of ***heat required for raising*** the temperature of ***unit mass*** of a substance ***through unit degree***.

- It is generally denoted by "C".
- The unit is KJ/kg. K

Specific heat is of two types,

- **Specific heat at constant pressure, Cp**
- **Specific heat at constant volume, $C_v$**

*The ratio of specific heats $C_p$ and Cv for any gas is assumed to be constant.*

This is expressed by the symbol " $\gamma$ ",

$$\gamma = C_p / Cv$$

- For air, $\gamma = 1.4$

Heat supplied to the gas at constant volume is

$Q = m\ C_v\ (T_2 - T_1)$ KJ

Heat supplied to the gas at constant pressure is

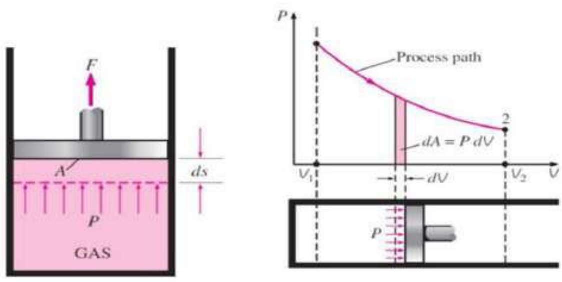
$Q = m\ Cp\ (T_2 - T_1)$ KJ



## WORK

- Work, Like heat is an energy interaction between a system and its surroundings.
- Work is the energy transfer associated with a force acting through a distance. A rising piston, a rotating shaft are all associated with work interactions.



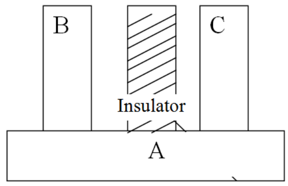



- For a closed system, Work, $\delta W$ or $\mathbf{W_{1\text{-}2}} = \int_1^2 \mathbf{PdV}$, area under the curve 1-2
- Unit , N-m or Joules(J)

# Laws of Thermodynamics

## Zeroth Law of Thermodynamics

- When a body is in thermal equilibrium with a body B, and also separately with a body C, then B and C will be in thermal equilibrium with each other.




# First Law of Thermodynamics

- According to this Law, when a closed system undergoes a Thermodynamic Cycle, the net Heat transfer is equal to net work transfer.

  §dQ = §dW

- The energy can be neither created nor destroyed though it can be transformed from one form to another
- For a process dQ=dU+dW



# Second Law of Thermodynamics

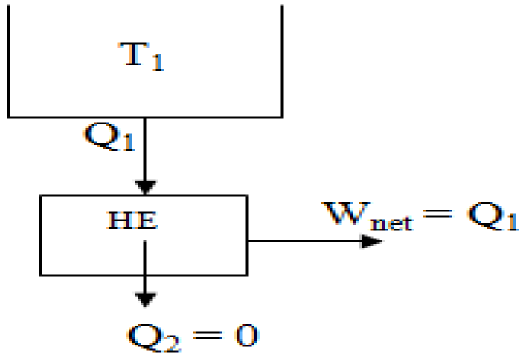
It has two common statements

- **Kelvin-Planck statement** and
- **Clausius statement**

## Kelvin-Planck statement

"It is impossible to construct an engine working on a cyclic process, whose sole purpose is to convert heat energy from a single thermal reservoir into an equivalent amount of work"

## *This type of engine is impossible*






## Clausius statement

"It is impossible for a self acting machine working on a cyclic process, to transfer heat from a body at a lower temperature to a body at higher temperature with out the aid of an external agency".



## Entropy

- **Entropy is defined as a measure of the degradation which energy experiences as a result of energy conversion.**
- It is also defined as the measure of irreversibility associated with any process.
- The more the irreversibility, the more will be the change in entropy.
- ***Entropy is a property, which cannot be measured directly, but the change of entropy during any process can be calculated.***

- Small change in entropy, dS is defined as the ratio of small amount of heat transfer, dQ, to the absolute temperature, T, at which heat is transferred.

ie, $$\mathbf{dS = \frac{dQ}{T}}$$



## Enthalpy

- Enthalpy is defined as the sum of internal energy and pressure volume product (flow work).

- It is denoted by H.

$$\mathbf{H = U + P\,V}$$

*Where, U is internal energy*

*P is pressure and*

*V is volume*



Change in enthalpy dH is expressed as,

$$\begin{aligned}
dH &= dU + d(p\,V) \\
H_2\text{-}H_1 &= (U_2 - U_1) + (p_2v_2 - p_1v_1) \\
&= mc_v\,(T_2 - T_1) + (mRT_2 - mRT_1) \\
&= mc_v\,(T_2 - T_1) + mR(T_2 - T_1) \\
&= m\,(T_2 - T_1)\,(c_v + R) \\
H_2\text{-}H_1 &= mc_p\,(T_2\text{-}T_1)
\end{aligned}
\quad \left\{ \begin{aligned} &\text{since } c_p - c_v = R \\ &c_p = c_v + R \end{aligned} \right.$$

$$\boxed{\Delta H = mc_p\,\Delta T}$$

## Thermodynamic Processes

- When a system changes its state from one equilibrium condition to another it is said to have undergone a process.

- When a system undergoes a thermodynamic process, the variation properties of the gas such as pressure, temperature, energy , entropy etc may change.



Thermodynamics process may be performed in different ways. Some of them are :

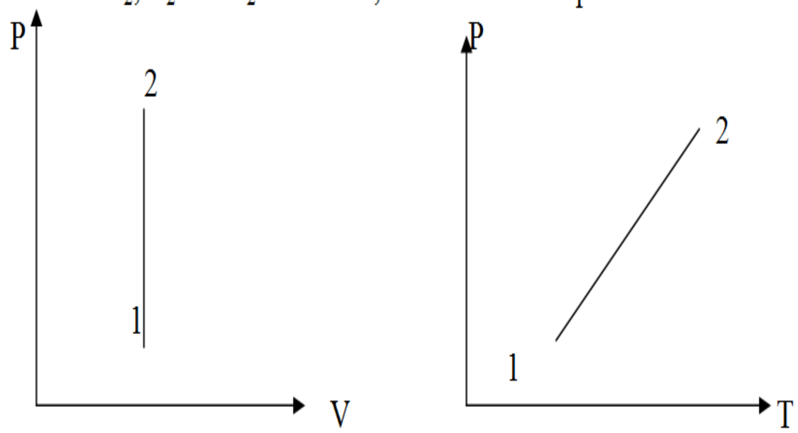
- Constant Volume Process or Isochoric Process
- Constant Pressure Process or Isobaric Process
- Constant Temperature Process or Isothermal Process
- Adiabatic Process
- Polytropic Process



# Constant Volume Process or Isochoric Process

- Consider 'm' kg of a gas being heated in a cylinder at constant volume from an initial temperature $T_1$ to final temperature $T_2$.
- Since there is no change in volume, no external work is done by the gas.
- The entire heat supplied will be stored in the form of internal energy.

$P_1$, $V_1$ and $T_1$ =Pressure, volume and Temperature at state1
$P_2$, $V_2$ and $T_2$ =Pressure, volume and Temperature at state2






## P-V-T relationship

For a Perfect Gas,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$V_1 = V_2$ ( since this is a constant volume process)

So, $$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$



## Work Done

**Work Done**

Work done $\delta W = p \times dV$

$W_{1-2} = p\,(v_1 - v_2)$

$= 0$ as $v_1 = v_2$

**Change in Internal Energy**

$dU = m \times C_v \times dT$

$$\int_1^2 dU = m\,c_v \int_1^2 dT$$

$U_2 - U_1 = m\,C_v\,(T_2 - T_1)$

## Heat Supplied

$$\delta Q = dU + \delta W$$

$$Q_{1-2} = m\,C_v\,(T_2 - T_1) + p\,(v_1 - v_2)$$

$$= m\,C_v\,(T_2 - T_1) + 0$$

$$\boxed{Q_{1-2} = m\,C_v\,(T_2 - T_1)}$$



# ENTHALPY

$$dH = dU + d(pv)$$

$$H_2 - H_1 = U_2 - U_1 + p_2v_2 - p_1v_1$$

Change in internal energy,

$$U_2 - U_1 = m\,C_v\,(T_2 - T_1)$$

$$H_2 - H_1 = m\,C_v\,(T_2 - T_1) + p_2v_2 - p_1v_1 \quad \text{---------------} \quad (1)$$

But
$p_1v_1 = mRT_1$ and
$p_2v_2 = mRT_2$

replacing $P_1V_1$ and $P_2V_2$ with $mRT_1$ and $mRT_2$ in equation (1), we get

$$H_2 - H_1 = m\,C_v\,(T_2 - T_1) + mRT_2 - mRT_1$$

$$= m\,C_v\,(T_2 - T_1) + mR(T_2 - T_1)$$

$$= m\,(T_2 - T_1)\,(C_v + R)$$

But $C_p - C_v = R$ and so

$$c_p = c_v + R$$

$$H_2 - H_1 = mc_p\,(T_2 - T_1)$$

$$\Delta H = mc_p\,\Delta T$$



# FOR ISOCHORIC PROCESS

- **P - V – T RELATIONSHIP**

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- **WORK DONE**

$$W = 0$$

- **INTERNAL ENERGY**

$$U_2 - U_1 = m\,C_v\,(T_2 - T_1)$$

- **HEAT SUPPLIED**

$$Q_{1-2} = m\,C_v\,(T_2 - T_1)$$

- **ENTHALPY**

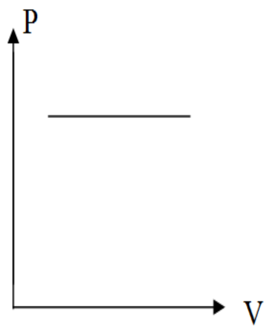
$$H_2 - H_1 = m\,C_p(T_2 - T_1)$$

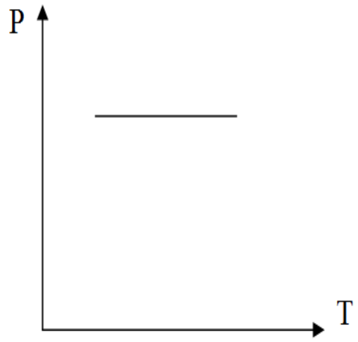
# CONSTANT PRESSUE PROCESS
# or
# Isobaric process

- When a gas is heated at constant pressure, its volume & temperature will increase.
- Heat supplied to the gas is utilized for increasing internal energy & for doing some external work.



Consider heating of m kg of a gas at constant pr. from state 1 to state 2.






## P-V-T RELATION

- $$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Since, $P_1 = P_2$ ( process is isobaric)

So, $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

## Work done

Work done by the gas

$$dW = PdV$$

*In Isochoric process, p = $p_1 = p_2$*

$$= p_2 V_2 - p_1 V_1$$

$$= P( V_2 - V_1 )$$

$$\mathbf{dW = mR(T_2 – T_1)}$$

## Change in internal energy

$$dU = mC_v dT$$

$$\mathbf{dU = mC_v(T_2 – T_1)}$$



## Heat Supplied

$$dQ = dU + dW$$

$$= U_2 – U_1 + PdV$$

$$= U_2 – U_1 + P( V_2 - V_1)$$

In Isochoric process, $p = p_1 = p_2$

$$= U_2 – U_1 + p_2 V_2 - p_1 V_1$$

$$dQ = mC_v(T_2 – T_1) + mR(T_2 – T_1)$$

$$= m (T_2 – T_1) (C_v + R) \qquad c_p – c_v = R$$

$$\mathbf{dQ = mC_p(T_2 – T_1)}$$



## Change in enthalpy

$$H_2 – H_1 = dU + dW$$

From previous derivations,

$$= mC_v(T_2 – T_1) + mR(T_2 – T_1)$$

$$= m (T_2 – T_1) (C_v + R)$$

$$\mathbf{H_2 – H_1 = m\, C_p ( T_2 - T_1 )}$$

## FOR ISOBARIC PROCESS

- **P-V-T RELATION**

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- **Work done**

$$dW = mR(T_2 - T_1)$$

- **Change in internal energy**

$$dU = mC_v(T_2 - T_1)$$

- **Heat Supplied**

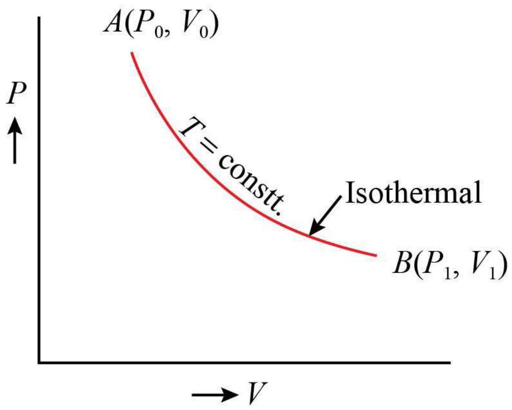
$$dQ = mC_p(T_2 - T_1)$$

- **Change in enthalpy**

$$H_2 - H_1 = m\,C_p\,(T_2 - T_1)$$



# ISOTHERMAL PROCESS




($P$–$V$ Diagram)



- **P-V-T RELATION**

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

For isothermal process, $T_1 = T_2$

$$\boxed{P_1V_1 = P_2V_2}$$

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

# WORK DONE

$$W_{1-2} = {}^{2}\int_{1} PdV \quad \text{----------------} \quad (i)$$

For an isothermal process, $pV = p_1V_1 = p_2V_2 = \text{Constant}$

or $$p = \frac{p_1V_1}{V}$$

Substituting this value of 'p' in eq. (i)

$${}_1W_2 = \int_{V_1}^{V_2} \frac{p_1V_1}{V} dV$$

$$= p_1V_1 \int_{V_1}^{V_2} \frac{dV}{V}$$



$${}_1W_2 = p_1V_1 [\ln V]_{V_1}^{V_2}$$

$$= p_1V_1 (\ln V_2 - \ln V_1)$$

$${}_1W_2 = p_1V_1 \ln \left\{\frac{V_2}{V_1}\right\}$$

Also for an isothermal process

$$p_1V_1 = p_2V_2$$

$$\therefore \frac{V_2}{V_1} = \frac{p_1}{p_2}$$

Substituting this,

$${}_1W_2 = p_1V_1 \ln \left\{\frac{p_1}{p_2}\right\}$$



## Change in internal energy

$$dU = mC_v dT$$

Since, $T_2 = T_1$, $dT = 0$

$$dU = 0$$

## Heat Supplied

$$dQ = dU + dW$$

Since dU= 0

$$dQ = dW$$

$$dQ = {}_1W_2 = p_1V_1 \ln\left\{\frac{V_2}{V_1}\right\}$$

or

$$_1W_2 = p_1V_1 \ln\left\{\frac{p_1}{p_2}\right\}$$



# ADIABATIC PROCESS

For an adiabatic process 1 - 2,

$$p_1V_1^{\gamma} = p_2V_2^{\gamma} = \text{constant}$$

1. $\dfrac{p_1}{p_2} = \left\{\dfrac{V_2}{V_1}\right\}^{\gamma}$

2. $\dfrac{p_1}{p_2} = \left\{\dfrac{T_1}{T_2}\right\}^{\frac{\gamma}{\gamma-1}}$

3. $\dfrac{V_2}{V_1} = \left\{\dfrac{T_1}{T_2}\right\}^{\frac{1}{\gamma-1}}$



# WORKDONE

$$_1W_2 = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$$

$$_1W_2 = \frac{mR\,(T_1 - T_2)}{(\gamma - 1)}$$

# CHANGE IN INTERNAL ENERGY

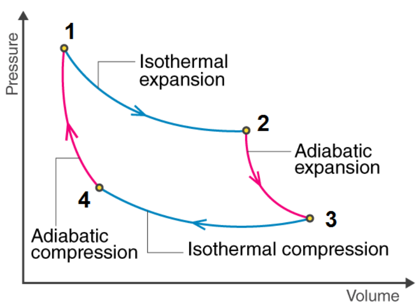
$$\therefore \Delta U = -\,{}_1W_2 = -\frac{mR\,(T_1 - T_2)}{(\gamma - 1)}$$

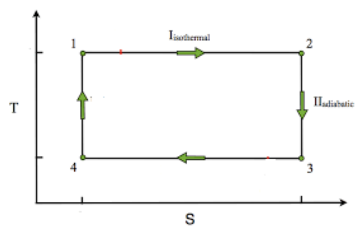
$$\boxed{\Delta U = \frac{mR\,(T_2 - T_1)}{(\gamma - 1)}}$$



| Process | p-V-T relation | Work-done | Internal energy | Heat supplied | Change in enthalpy |
|---|---|---|---|---|---|
| Isochoric | $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ | 0 | $mc_v(T_2 - T_1)$ | $mc_v(T_2 - T_1)$ | $mc_p(T_2 - T_1)$ |
| Isobaric | $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ | $p(V_2 - V_1)$ | $mc_v(T_2 - T_1)$ | $mc_p(T_2 - T_1)$ | $mc_p(T_2 - T_1)$ |
| Isothermal | $p_1V_1 = p_2V_2$ | $p_1V_1 \log_e\left(\frac{V_2}{V_1}\right)$ | 0 | $p_1V_1 \log_e\left(\frac{V_2}{V_1}\right)$ | 0 |
| Adiabatic | $\frac{p_1}{p_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$ <br> $\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$ <br> $\frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{\frac{\gamma-1}{\gamma}}$ | $\frac{p_1V_1 - p_2V_2}{\gamma - 1}$ | $mc_v(T_2 - T_1)$ | 0 | $mc_p(T_2 - T_1)$ |
| Polytropic | $\frac{p_1}{p_2} = \left(\frac{V_2}{V_1}\right)^{n}$ <br> $\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{n-1}$ <br> $\frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{\frac{n-1}{n}}$ | $\frac{p_1V_1 - p_2V_2}{n - 1}$ | $mc_v(T_2 - T_1)$ | $\frac{p_1V_1 - p_2V_2}{n - 1} + mc_v(T_2 - T_1)$ | $mc_p(T_2 - T_1)$ |

# CARNOT CYCLE



Efficiency $\eta = \dfrac{\text{work done}}{\text{Heat supplied}}$

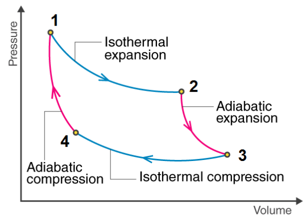
work done $=$ Heat supplied $-$ Heat Rejected

$\therefore \quad \eta = \dfrac{\text{Heat supplied} - \text{Heat Rejected}}{\text{Heat supplied}}$

or

$$\eta = 1 - \frac{\text{Heat Rejected}}{\text{Heat supplied}} \longrightarrow (1)$$






For Carnot Cycle

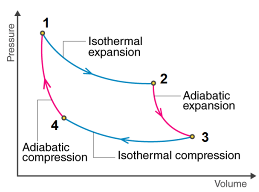
Heat supplied is during process 1–2 and is isothermal process.

$$H_{S_{1-2}} = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

or

$$= mRT_1 \ln\left(\frac{V_2}{V_1}\right) \rightarrow (2)$$






Heat Rejected is during process 3–4 and is isothermal.

$$H_{R_{3-4}} = P_3 V_3 \ln\left(\frac{V_3}{V_4}\right)$$

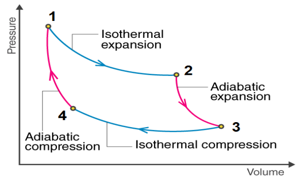
or

$$= mRT_3 \ln\left(\frac{V_3}{V_4}\right) \longrightarrow (3)$$

Subs: (3) & (4) in (1)

$$\eta = \frac{H_S - H_R}{H_S}$$

$$= \frac{mRT_1 \ln\left(\frac{V_2}{V_1}\right) - mRT_3 \ln\left(\frac{V_3}{V_4}\right)}{mRT_1 \ln\left(\frac{V_2}{V_1}\right)}$$




Consider process 2 - 3, [adiabatic expansion]

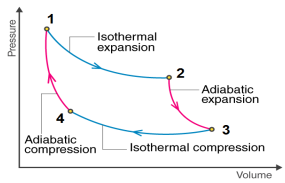
by PVT relation,

$$\frac{V_3}{V_2} = \left(\frac{T_2}{T_3}\right)^{\frac{1}{\gamma-1}} \longrightarrow (5)$$

for process 4 - 1,

$$\frac{V_4}{V_1} = \left(\frac{T_1}{T_4}\right)^{\frac{1}{\gamma-1}} \longrightarrow (6)$$






but considering process 1 - 2, and 3 - 4 isothermal process,

$T_1 = T_2$ and

$T_3 = T_4$.

Subs, $T_2$ as $T_1$ and $T_3$ as $T_4$ in eqn (5)

$$\frac{V_3}{V_2} = \left(\frac{T_1}{T_4}\right)^{\frac{1}{\gamma-1}} \longrightarrow (7)$$

Considering eqn (6) & (7) since RHS are same,

$$\frac{V_3}{V_2} = \frac{V_4}{V_1}$$

$$\therefore \frac{V_3}{V_4} = \frac{V_2}{V_1} \longrightarrow (8)$$



Substituting $\frac{V_3}{V_4} = \frac{V_2}{V_1}$ (as in eqn (8)) in Eqn (4)

$$\eta = \frac{mRT_1 \ln\left(\frac{V_2}{V_1}\right) - mRT_3 \ln\left(\frac{V_2}{V_1}\right)}{mR\,T_1 \ln\left(\frac{V_2}{V_1}\right)}$$

$$\eta = \frac{mR \ln\left(\frac{V_2}{V_1}\right)\left[T_1 - T_3\right]}{mR \ln\left(\frac{V_2}{V_1}\right) \cdot T_1}$$

$$\eta = \frac{T_1 - T_3}{T_1}$$

$$\eta = \frac{T_1 - T_3}{T_1}$$

$$\eta = 1 - \frac{T_3}{T_1}$$

ie,
$$\boxed{\eta = 1 - \frac{\text{Temperature of Cold body}}{\text{Temperature of Hot body}}}$$

Normally, Temp of hot body is taken as $T_1$ and Cold body as $T_2$.

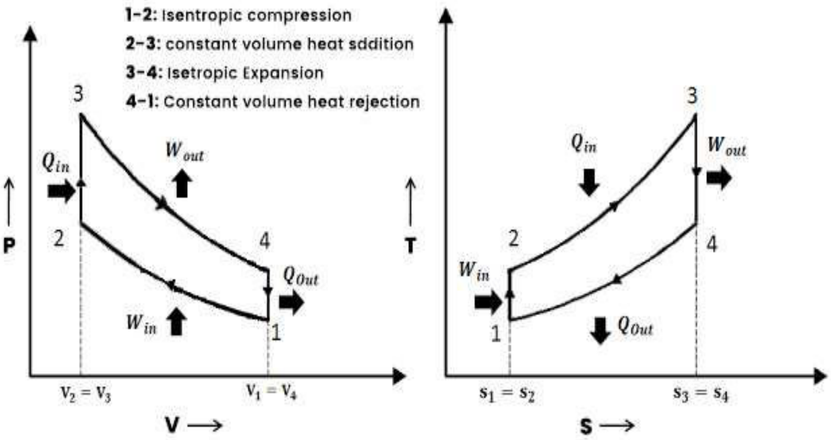
$$\therefore \boxed{\eta = 1 - \frac{T_2}{T_1}}$$



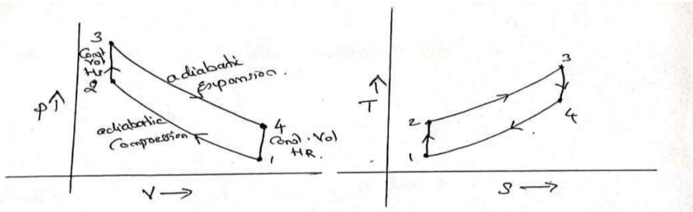


# OTTO CYCLE

**( CONSTANT VOLUME CYCLE)**




**P-V and T-S Diagram of Otto Cycle**




$$\eta = 1 - \frac{HR}{HS}$$

Heat is supplied during process 2-3 @ Const. Vol.

$$\therefore \quad HS = m C_v (T_3 - T_2)$$

Heat is rejected during process 4-1 at Const. Vol.

$$\therefore \quad HR = m C_v (T_4 - T_1)$$

$$\therefore \quad \eta = 1 - \frac{m C_v (T_4 - T_1)}{m C_v (T_3 - T_2)} \longrightarrow (1)$$

Consider process 1-2 (adiabatic process)

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

but $\frac{V_1}{V_2}$ is called Compression ratio 'r'

$$\therefore \frac{T_2}{T_1} = r^{\gamma-1}$$

$$\therefore T_2 = T_1 \cdot r^{\gamma-1} \longrightarrow (2)$$



Consider process 3 – 4 (adiabatic process)

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1} \longrightarrow (3)$$

but 2 - 3 and 4 – 1 are Const. Vol. process.

$\therefore V_3 = V_2$ and $V_4 = V_1$ } Subs in eqn (3)

$$\therefore \frac{T_3}{T_4} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}, \text{ but } \frac{V_1}{V_2} = r$$

$$\therefore \frac{T_3}{T_4} = r^{\gamma-1}$$

$$T_3 = T_4 \cdot r^{\gamma-1} \longrightarrow (4)$$

Subs (2) & (4) in (1)



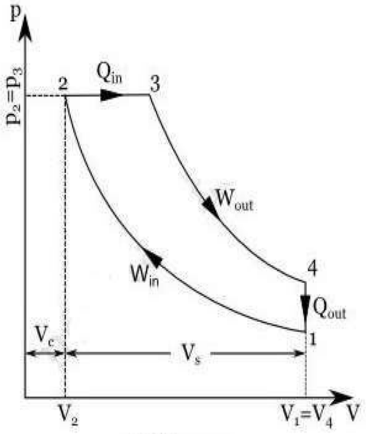
$$\eta = 1 - \frac{(T_4 - T_1)}{T_4 \cdot r^{\gamma-1} - T_1 \cdot r^{\gamma-1}}$$

$$= 1 - \frac{(T_4 - T_1)}{r^{\gamma-1}(T_4 - T_1)}$$

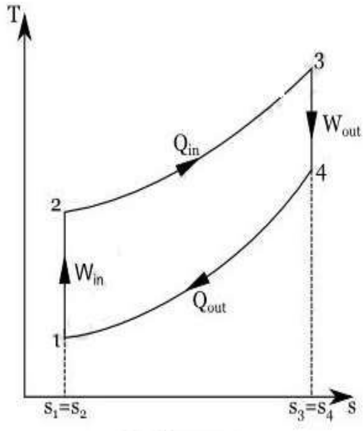
$$\boxed{\therefore \eta = 1 - \frac{1}{r^{\gamma-1}}}$$

# DIESEL CYCLE

( CONSTANT PRESSURE CYCLE)




p-V Diagram




T-s Diagram



$$\eta = 1 - \frac{H_R}{H_S}$$

Heat is supplied at constant pressure during process 2 – 3.

$$\therefore \quad H_S = m C_p (T_3 - T_2)$$

Heat is rejected at constant Volume during process 4 – 1

$$\therefore \quad H_R = m C_v (T_4 - T_1)$$

$$\therefore \eta = 1 - \frac{m C_v (T_4 - T_1)}{m C_p (T_3 - T_2)}$$

Since $\frac{C_p}{C_v} = \gamma$.

$$\eta = 1 - \frac{(T_4 - T_1)}{\gamma \cdot (T_3 - T_2)} \rightarrow (1)$$

$\frac{V_3}{V_2}$ is called Cutoff ratio and is represented by '$\rho$'

$\frac{V_4}{V_3}$ is called expansion ratio and is represented by '$r_1$'

$$\frac{V_4}{V_3} = \frac{V_4}{V_2} \times \frac{V_2}{V_3}$$

but process 4 – 1 is Constant Volume, and $V_4 = V_1$

$$\frac{V_4}{V_3} = \frac{V_1}{V_2} \times \frac{V_2}{V_3}$$

ie, $\quad r_1 = r \times \frac{1}{\rho}$

or $\quad r_1 = \frac{r}{\rho}$,

$r_1$ is expansion ratio
$r$ is Compression ratio
$\rho$ is cutoff ratio.

Consider process 1 - 2 (adiabatic process)

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma - 1}$$

but $\frac{V_1}{V_2} = r$, Compression ratio.

$$\therefore \frac{T_2}{T_1} = r^{\gamma - 1}$$

$$\therefore T_2 = T_1 \cdot r^{\gamma - 1} \longrightarrow (2)$$



Consider process 2 - 3 (const. pressure process)

$$\frac{T_3}{T_2} = \frac{V_3}{V_2}$$

$$\frac{T_3}{T_2} = \rho$$

$$\therefore T_3 = T_2 \cdot \rho.$$

Subs. Value of $T_2$ from eqn (2)

$$T_3 = T_1 \cdot r^{\gamma - 1} \cdot \rho \longrightarrow (3)$$



Consider process 3 - 4 (adiabatic process)

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma - 1}$$

ie, $\frac{T_3}{T_4} = r_1^{\gamma - 1}$

{ Since $r_1 = \frac{r}{\rho}$

$$\therefore \frac{T_3}{T_4} = \frac{r^{\gamma - 1}}{\rho^{\gamma - 1}}$$

$$\therefore T_4 = T_3 \cdot \frac{\rho^{\gamma - 1}}{r^{\gamma - 1}}$$

Subs Value of $T_3$ as in eqn (3)

$$T_4 = T_1 \cdot r^{\gamma - 1} \cdot \rho \cdot \frac{\rho^{\gamma - 1}}{r^{\gamma - 1}}$$

$$\therefore \quad T_4 = T_1 \rho^{\gamma} \longrightarrow (4)$$

Subs eqn (2), (3) and (4) in eqn (1)

$$\eta = 1 - \frac{(T_4 - T_1)}{\gamma (T_3 - T_2)}$$

$$\therefore \quad \eta = 1 - \frac{\left(T_1 \cdot \rho^{\gamma} - T_1\right)}{\gamma \cdot \left(T_1 \cdot r^{\gamma-1} \cdot \rho - T_1 \cdot r^{\gamma-1}\right)}$$

$$1 - \frac{T_1 \left(\rho^{\gamma} - 1\right)}{\gamma \cdot T_1 \cdot r^{\gamma-1} (\rho - 1)}$$

$$\therefore \eta = 1 - \frac{\left(\rho^{\gamma} - 1\right)}{\gamma \cdot r^{\gamma-1} (\rho - 1)}$$

$$\therefore \eta = 1 - \frac{1}{r^{\gamma-1}} \cdot \frac{1}{\gamma} \cdot \frac{\left(\rho^{\gamma} - 1\right)}{(\rho - 1)}$$

$$\boxed{\therefore \eta = 1 - \frac{1}{r^{\gamma-1}} \cdot \frac{1}{\gamma} \left[\frac{\rho^{\gamma} - 1}{\rho - 1}\right]}$$